

From bathroom inspiration to an explained, buildable design.

KOHLER Spatia connects user intent, spatial constraints, product compatibility, sustainability and conversational AI in one continuous workflow.

THE PROBLEM

Bathroom planning is fragmented.

Users must balance dimensions, budget, style, fixtures, accessibility and sustainability across disconnected decisions.

THE SPATIA APPROACH

- 01 Define** Space, budget, accessibility, priorities and theme.
- 02 Configure** Select fixtures with fit and style feedback.
- 03 Reason** Expose AI recommendations, conflicts and resolutions.
- 04 Visualize** Move from plan to architectural dollhouse view.
- 05 Decide** Compare sustainability, ask the designer and export specs.

CORE DESIGN PRINCIPLES

Constraint-aware

Physical fit and design conflicts are surfaced instead of hidden.

Explainable

AI Confidence, Fit Score and AI Reasoning Studio make recommendations inspectable.

CORE IDEA

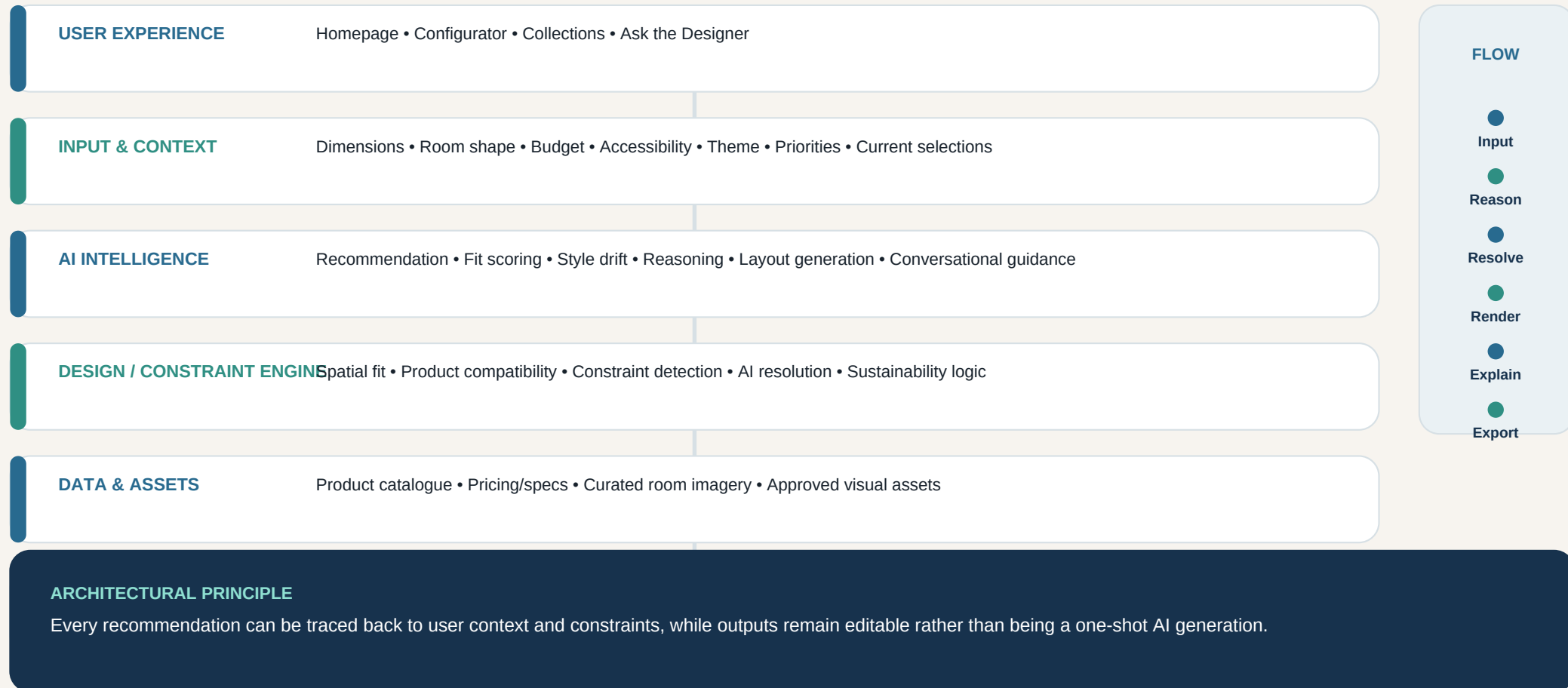
AI acts as a design co-pilot: recommend → explain → resolve → visualize → document.

End-to-end

The workflow continues from inspiration to visual output and specification report.

A layered pipeline keeps design decisions connected.

The architecture separates the experience layer from AI reasoning, constraint resolution, product data and presentation outputs.



A modern web stack for interactive, visual and AI-assisted planning.

The implementation prioritizes a fast browser experience, reusable UI, interactive spatial visualization and explainable design interactions.

FRONTEND

React
TypeScript
Vite

Component-driven application architecture and fast development.

UI / STYLING

Tailwind CSS
Responsive UI
Design system

Consistent premium interface with reusable visual patterns.

3D / SPATIAL

Three.js
2D / 3D views
Dollhouse cutaway

Interactive representation of the configured bathroom.

AI LAYER

Recommendation logic
Reasoning + chat
Constraint resolution

AI-assisted selection, explanation, conflict handling and conversation.

DATA / ASSETS

Product catalogue
Pricing + specs
Curated imagery

Structured product information and approved visual assets.

OUTPUT / ANALYTICS

Design metrics
Comparison graphs
Specification PDF

Communicate decisions through visual metrics and a portable report.

IMPLEMENTATION FOCUS Responsive • Explainable AI • Interactive visualization • Reusable components • Exportable outputs

Spatia turns AI from a generator into a design co-pilot.

The innovation is the connection between decision layers throughout the bathroom planning journey—not one isolated AI feature.

THE PITCH

What if a bathroom designer could explain every recommendation, catch physical conflicts, and still leave the final decision with the user?

01

Explainable Design

AI Confidence, Fit Score, Style Drift and AI Reasoning Studio make recommendations understandable.

02

Constraint-Aware AI

Dimensions, compatibility, budget and accessibility become active design constraints.

03

Multiple Entry Points

Start from configuration, collections, conversation, photos, life-stage planning or sustainability.

04

Design → Decision

End with visual validation, sustainability metrics and a practical specification report.

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